



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 • STANDARD 6.AT.3

# Ratio and Rate Problem Solving

Lesson 3-7 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_

## My Notes

Level 3 Enrichment



### TODAY'S OBJECTIVES

**Content Objective:** I can solve real-world rate problems by finding and using unit rates.

**Language Objective:** I can explain my solution using the words rate, unit rate, per, and proportion.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Rate

Unit rate

Per

Problem solving

Proportion



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

- 1 A ratio comparing two quantities with different units is a

.

- 2 A rate for exactly one unit, like \$3 per 1 pound, is a  .

- 3 The word that means 'for each' in a rate, like miles  hour, is  .

- 4 Using a plan and math steps to find an answer to a real situation is

.

- 5 An equation that two ratios are equal is a  .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Team 2 frosts 5 cupcakes every 4 minutes. How did you set up a rate or proportion to find how many cupcakes they frost in a longer time?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **Rate** — A ratio comparing two amounts with different units, like miles per hour.  
*Tasa — Una razón que compara dos cantidades con unidades distintas, como millas por hora.*

☐ **Unit rate** — A rate for just 1 of something, like cost for 1 item.  
*Tasa unitaria — Una tasa para solo 1 de algo, como el precio de 1 artículo.*

☐ **Per** — For each one. Example: 5 dollars per book.  
*Por — Por cada uno. Ejemplo: 5 dólares por libro.*

☐ **Proportion** — A math sentence saying two ratios are equal. It helps find a missing number.  
*Proporción — Una oración matemática que dice que dos razones son iguales. Ayuda a hallar un número que falta.*

☐ **Problem solving** — Using ratios and rates to find a missing amount.  
*Resolución de problemas — Usar razones y tasas para encontrar una cantidad que falta.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The rate is \_\_\_\_ cupcakes per \_\_\_\_ minutes.**

*La tasa es \_\_\_\_ cupcakes por \_\_\_\_ minutos.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** Divide the total cost by the number of pounds to find the cost per 1 pound.

**Evidence:** Students divide \$18 by 3 to get \$6 per pound and explain the unit rate lets them find the cost of any number of pounds.

**Reasoning:** This evidence connects the definition of rate to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The rate is \_\_\_\_ cupcakes per \_\_\_\_ minutes.**

*La tasa es \_\_\_\_ cupcakes por \_\_\_\_ minutos.*

- **To find the total, I \_\_\_\_ the rate.**

*Para hallar el total, \_\_\_\_ la tasa.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**

*Mi afirmación es \_\_\_\_.*

- **I know because \_\_\_\_.**

*Lo sé porque \_\_\_\_.*

- **So \_\_\_\_.**

*Entonces \_\_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Rate
2. **Notes 2:** Unit rate
3. **Notes 3:** Per
4. **Notes 4:** Problem solving
5. **Notes 5:** Proportion
6. **Watch for this mistake:** *A common mistake in Ratio and Rate Problem Solving is dividing in the wrong order when setting up the unit rate — for example, finding "minutes per smoothie" ( $6 \div 18$ ) instead of "smoothies per minute" ( $18 \div 6$ ) — or using addition instead of multiplication to scale the unit rate, like adding  $\$0.75 + 10$  instead of multiplying  $\$0.75 \times 10$ . Before you submit, check: does your unit rate match the units the question is asking for, and did you multiply (not add) to scale it to the new amount?*
7. **Try It 1:** A. 28 carrots — *Unit rate:  $8 \text{ carrots} \div 2 \text{ minutes} = 4 \text{ carrots per minute}$ . In 7 minutes:  $4 \times 7 = 28 \text{ carrots}$ .*
8. **Try It 2:** A. 42 ounces — *Unit rate:  $18 \text{ ounces} \div 3 \text{ loaves} = 6 \text{ ounces per loaf}$ . For 7 loaves:  $6 \times 7 = 42 \text{ ounces}$ . The equivalent ratio  $18/3 = 42/7$  confirms it.*